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1 Introduction

2 Related Work

2.1 Limitations of the Base Study

While Go et al. [1] demonstrated the feasibility of an automated system for identifying and classifying viable pollen grains, their proposed image-analysis protocol encountered difficulties when pollen grains overlapped or occurred in clustered ("lumped") arrangements. The method used edge detection, dilation, region filling, and k-means clustering, and relied on manually selected image-processing parameters, limiting its adaptability to variations in pollen images. In particular, accuracy decreased from 98.3% for images without lumped pollen to 85.06% for images containing lumped pollen because the edge-detection process failed to consistently distinguish overlapping grains. These limitations highlighted the need for more adaptive machine-learning-based methods. Subsequent studies have explored machine learning, deep learning, segmentation, and transfer-learning techniques to address these limitations, providing a basis for determining whether the identified gap has been addressed, partially addressed, or remains unresolved.

2.2 Identification and Detection of Pollen Grains

Recent studies have increasingly relied on machine learning and deep learning to reduce the limitations of manually designed image-processing procedures. Rather than depending on predefined parameters, as in Go et al. [1], these approaches attempt to learn useful pollen features directly from image data.

Viertel and König [2] reviewed automated pollen-image classification methods and compared traditional machine-learning approaches with convolutional neural networks (CNNs). Their review highlighted the increasing use of deep learning while also identifying continuing challenges involving image acquisition, dataset diversity, and model generalization. Chen and Ju [3] addressed image-quality variation by combining SRGAN-based deblurring with a multi-scale classifier. Their ResNet-50-based model achieved 97.7% accuracy, demonstrating that improving the quality of microscopic images before classification can improve recognition performance. Jin et al. [4] further extended automated pollen detection to partially labeled datasets through Faster R-CNN and a teacher-student semi-supervised learning strategy. Their approach achieved an F1 score of 76.9%, compared with 61.3% for the BAA500 automated pollen monitor, showing that machine learning can improve detection even when complete manual annotation is unavailable. Shamrat et al. [5] also developed PollenNet,

which combined deep learning with preprocessing procedures such as removal of low-quality and duplicate images, noise reduction, brightness adjustment, and explainable artificial intelligence.

These studies demonstrate a progression from fixed image-processing parameters toward models that learn features from data and can accommodate different image conditions. However, image quality, incomplete labeling, and dataset diversity continue to influence detection performance.

2.3 Detection and Segmentation of Overlapping or Clustered Pollen Grains

The difficulty of identifying lumped pollen grains is the most direct limitation of Go et al. [1]. Because dilation and region-filling operations depended on correctly detected pollen boundaries, failures during detection affected the final pollen count and viability classification. This limitation has encouraged researchers to use segmentation and object-detection approaches that can distinguish individual objects within complex images.

Li et al. [6] proposed a weakly supervised collaborative-learning framework that combined pollen detection, U-Net segmentation, and CNN classification. The method generated segmentation information from pollen contours and allowed segmentation and classification to improve each other. Using approximately 1,000 images with image-level labels, the method achieved 86.6% pollen-identification accuracy, demonstrating that segmentation can reduce interference from impurities while also reducing the need for extensive pixel-level annotation.

Zhang et al. [7] used Mask R-CNN instance segmentation to detect pollen germination and pollen-tube length from microscopic images. To improve recognition of difficult structures, synthetic images containing crossed pollen tubes were added to the training data. The model achieved an mAP of 0.949 at an IoU threshold of 0.50, while its measurements of germination frequency and pollen-tube length produced R^2 values of 0.909 and 0.958, respectively. The model was also successfully applied to two additional plant species, indicating some capacity for generalization.

Gallardo et al. [8] approached the problem through multifocus microscopy combined with deep-learning object detection. Faster R-CNN and RetinaNet were used to process multiple focal planes, improving the detection of grains that might be blurred or obscured in individual images. The final system achieved a macro-F1 score of 0.956. However, highly overlapping grains and multiple detections across focal planes could complicate non-maximum suppression, showing that multifocus imaging can introduce its own detection challenges.

Feng et al. [9] further demonstrated that pollen segmentation can be affected by differences

between imaging domains. Their study investigated mixture-of-experts models, domain fine-tuning, continual learning, and foundation-model segmentation to detect and isolate pollen on palynological slides. Fine-tuning substantially improved detection when models were transferred between imaging domains, while segmentation models were able to further refine detected pollen boundaries. These findings show that improving segmentation alone may not be sufficient when a model is applied to substantially different datasets.

Overall, recent object-detection and segmentation techniques have substantially improved the handling of complex pollen arrangements compared with the edge-based method of Go et al. [1]. Nevertheless, overlapping grains, different imaging conditions, focal-plane variation, and insufficiently diverse training data remain challenges.

2.4 Machine Learning for Adaptive Pollen Viability Classification

Go et al. [1] classified viable and nonviable pollen using k-means clustering based on grayscale intensity after staining. Although effective for relatively separated grains, the process depended on the accuracy of the preceding image-processing stages. The authors consequently suggested that machine learning could make pollen identification and processing parameters more dynamic.

Recent studies directly address this limitation through deep learning. Tan et al. [10] developed PollenDetect, a modified YOLOv5 model designed to detect small pollen targets and distinguish viable from nonviable stained pollen. The system achieved 99% accuracy while reducing processing time from approximately three minutes to about one second per image. It also performed better than YOLOv3 and Faster R-CNN and showed improved detection of occluded and overlapping pollen aggregates. However, the system required additional training and parameter adjustment when applied to other plant species, illustrating a remaining transferability limitation.

Yamazaki et al. [11] approached viability through pollen germination rather than staining. Their YOLOv5 transfer-learning model distinguished germinated from non-germinated pollen and estimated germination rates across several plant species. Together with the germination measurements of Zhang et al. [7], this shows that machine learning can move beyond simple viable/nonviable classification and support more detailed quantitative measurements of pollen viability.

Other researchers have focused on making pollen classification more robust under difficult image conditions. Li et al. [12] developed a prior knowledge-guided deep feature-learning model that incorporated pollen location, shape, and texture information to guide CNN attention. The approach achieved accuracy above 88% on real-world pollen images, demonstrating the

value of combining domain-specific pollen characteristics with deep learning. Zolfaghari and Sajedi [13] similarly developed RCANet, which uses separate attention mechanisms for pollen structure and spatial information. Across three pollen datasets, the model achieved weighted F1 scores of 98.69%, 97.83%, and 98.24%.

The most recent work further targets challenging imaging conditions. Li et al. [14] developed PVDT, which combines YOLOv10-CBAM, pollen-image segmentation, and attention-based processing. The system specifically considered variable lighting, high pollen density, and low image quality, achieving 90–98% accuracy under these conditions. Transfer learning also enabled pollen-viability detection across ten additional plant species.

These studies indicate that machine learning has significantly advanced the adaptability of automated pollen analysis. However, differences in species, pollen density, image quality, and imaging environments can still affect performance.

2.5 Research Gap Analysis and Possible Research Direction

While subsequent studies have demonstrated that deep-learning object detection and instance segmentation can more reliably isolate complex and overlapping pollen grains [7], [8], [10], [14], these approaches have not been applied to or evaluated on rice pollen viability classification. Go et al. [1] remain the only reviewed study addressing automated viability identification of rice pollen, and their edge-based protocol lost accuracy when pollen grains overlapped or occurred in lumped arrangements. The deep-learning methods that better handle such arrangements were developed for other species or pollen types and frequently required additional training before being transferred to new crops [9], [10]. Furthermore, apart from Go et al. [1], the reviewed studies do not report viability performance separately for isolated and overlapping grains, leaving the effect of overlap on classification unquantified.

To address this gap, this study proposes to fine-tune a pretrained YOLO instance-segmentation model to detect individual rice pollen grains and classify each as viable or nonviable. A dataset of about 200 to 300 stained rice pollen images will be annotated at the grain level and grouped by overlap as isolated or lumped, with copy-paste augmentation used to generate additional overlapping examples. The model will be compared with the edge-based protocol of Go et al. [1] using precision, recall, and F1 score for each overlap group, and its viability percentages will be compared with manual pollen counts. Through this approach, the study aims to determine whether instance segmentation can reduce the drop in accuracy on overlapping rice pollen grains.

3 Problem Statement

Rice pollen viability is commonly assessed by manually counting stained grains under a microscope, which is a slow and labor-intensive process. Go et al. [1] automated this task using an edge-based image-analysis protocol, but accuracy decreased from 98.3% to 85.06% when pollen grains overlapped or clustered. Although deep-learning methods have improved the separation of overlapping pollen in other species [7], [10], [14], they have not been evaluated on rice pollen viability classification. Therefore, there remains a need for an automated method that can reliably classify rice pollen viability even when grains overlap.

Specifically, this study seeks to answer the following questions:

1. How accurately can a YOLO-based instance-segmentation model detect individual rice pollen grains and classify them as viable or nonviable?
2. How does the model's performance on isolated grains compare with its performance on lumped grains?
3. How does the model compare with the edge-based protocol of Go et al. [1], particularly on images containing lumped pollen?
4. How closely do the model's estimated viability percentages agree with manual pollen counts?

4 Objectives

5 Methodology

This study will follow six stages: data collection, annotation, dataset preparation, model training, baseline implementation, and evaluation. Each stage is designed to answer the research questions stated in Section 3.

5.1 Data Collection

The study will use the dataset of Go et al. [1], which consists of stained microscopic pollen images of six rice genotypes: IR68652-3B-8-1, Sadri, CSR28, IR67075-2B-5-2, Rc222, and Pokkali. The dataset is divided into images without lumped pollen and images with lumped pollen, allowing direct comparison with the results reported by Go et al. [1]. Permission to use the dataset will be requested from the original authors. If additional images are needed,

new samples will be collected from the same genotypes using the same staining and imaging procedure to maintain consistency with the original dataset.

5.2 Annotation

Each pollen grain in the dataset will be annotated by tracing its outline as a polygon and labeling it as either viable or nonviable based on its staining, using a free annotation tool such as Roboflow or CVAT. In images with lumped pollen, touching or overlapping grains will be traced individually so that each grain is treated as a separate object. The existing grouping of the dataset into images with and without lumped pollen will be retained for evaluation. The number of viable and nonviable grains in each image will also be recorded from the annotations and verified by an experienced evaluator to serve as the reference for computing viability percentages.

5.3 Dataset Preparation

Due to the limited number of images in the dataset, the annotated images will be divided into training (60%), validation (20%), and testing (20%) sets. The split will be stratified so that each set contains a proportional number of images with and without lumped pollen, as well as images from each genotype. If the testing set is too small to produce reliable results, stratified 5-fold cross-validation [15] will be used instead, and the results will be averaged across all folds.

To compensate for the small dataset, data augmentation will be applied to the training set only. Copy-paste augmentation [16] will be used to place annotated pollen grains onto other images so that they overlap, increasing the number of lumped examples. Photometric augmentation, including variations in brightness, contrast, and blur, will account for differences in staining and focus, while geometric augmentation such as flipping and rotation will account for differences in grain orientation. The validation and testing sets will not be augmented so that evaluation reflects the original images.

5.4 Model Training

A lightweight YOLOv8 nano instance-segmentation model (YOLOv8n-seg) [17], pretrained on the COCO dataset [18], will be fine-tuned to detect individual rice pollen grains, generate a segmentation mask for each grain, and classify each grain as viable or nonviable. YOLO models treat object detection as a single regression problem, predicting object locations and classes in one pass through the network, which makes them fast enough for practical use [19].

The nano variant was selected because it has the fewest parameters among the YOLOv8 models, allowing it to be trained with limited computing resources while reducing the risk of overfitting on a small dataset. Transfer learning [20] will be applied by freezing the backbone of the pretrained network and training only the remaining layers, which reduces both training time and memory usage.

Training will be performed using the Ultralytics framework on the free GPU tier of Google Colab. The model will be trained with an input size of 640 pixels, a batch size of 8, and a maximum of 50 epochs, using the default optimizer and learning-rate settings of the framework. Mixed-precision training will be used to further reduce memory consumption, and early stopping will end training if the validation performance does not improve for 10 consecutive epochs. The model weights with the best validation performance will then be used for evaluation on the testing set. Given the small size of the dataset, each training run is expected to take only a few minutes.

During inference, overlapping predictions for the same grain will be reduced to a single prediction using non-maximum suppression [21]. The overlap between two regions A and B is measured using the intersection over union (IoU) [22]:

$$\text{IoU} = \frac{\text{Area of overlap between } A \text{ and } B}{\text{Area of union of } A \text{ and } B} \tag{1}$$

The steps of the proposed method are shown in Algorithm 1.

Algorithm 1 Proposed pollen viability estimation

Require: Microscopic pollen image

Ensure: Viability percentage of the image

1: Input the image into the trained YOLOv8n-seg model
2: Obtain the mask and class (viable or nonviable) of each detected grain
3: Remove duplicate predictions using non-maximum suppression
4: Count the viable and nonviable grains
5: Compute the viability percentage using Eq. (5)

5.5 Baseline

To compare the proposed model with the base study, the image-analysis protocol of Go et al. [1] will be re-implemented using OpenCV, as outlined in Algorithm 2. Following the original configuration, the pipeline will apply Canny edge detection [23] with a threshold of 70 to find the boundaries of the pollen grains, dilation with a value of 1 to close small gaps in the detected edges, and region filling to fill the interior of each grain [24]. The detected

grains will then be classified using k-means clustering [25] with $k = 2$ on the average grayscale intensity of each grain, separating them into a darker viable group and a lighter nonviable group. The baseline will be evaluated on the same testing set as the proposed model so that both methods are compared under identical conditions.

Algorithm 2 Baseline pollen viability identification of Go et al. [1]

Require: Microscopic pollen image

Ensure: Viability percentage of the image

1: Apply Canny edge detection with a threshold of 70
2: Apply dilation with a value of 1
3: Apply region filling to obtain each pollen grain
4: Compute the average grayscale intensity of each grain
5: Group the grains into viable and nonviable using k-means with $k = 2$
6: Compute the viability percentage using Eq. (5)

5.6 Evaluation

The proposed model and the baseline will be evaluated on the testing set using the metrics summarized in Table 1. A predicted grain will be counted as a true positive (TP) if it overlaps an annotated grain with an IoU (Eq. (1)) of at least 0.5, following the PASCAL VOC criterion [22], and its predicted viability class matches the annotation. Incorrect predictions are counted as false positives (FP), and annotated grains that were not detected are counted as false negatives (FN). From these counts, precision, recall, and F1 score [26] will be computed for each class:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

All metrics will be reported separately for images with and without lumped pollen to measure the effect of overlap on performance.

The viability percentage of each image, following Go et al. [1], is computed as:

$$V = \frac{N_v}{N_v + N_n} \times 100 \tag{5}$$

where N_v and N_n are the numbers of viable and nonviable grains in the image. The estimated

viability percentages of both methods will be compared with the manual counts using the mean absolute error (MAE) [27]:

$$\text{MAE} = \frac{1}{n} \sum_{j=1}^{n} \left| \hat{V}_j - V_j \right| \tag{6}$$

where n is the number of testing images, $\hat{V}_j$ is the estimated viability percentage, and V_j is the manual viability percentage of image j. A lower MAE indicates closer agreement with manual counting. The Pearson correlation coefficient [28] will also be computed to measure how consistently the estimated percentages follow the manual counts, where a value closer to 1 indicates stronger agreement.

Table 1: Evaluation metrics for each research question

Question	**Purpose**	**Metrics**
RQ1	Detection and viability classification	Precision, recall, and F1 score per class
RQ2	Effect of overlap	Metrics computed separately for images with and without lumped pollen
RQ3	Comparison with Go et al. [1]	Metrics of both methods for each overlap group
RQ4	Agreement with manual counts	MAE and Pearson correlation of viability percentages

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